BASE CHANGE OF CYCLES AND OF THE K-CYCLE OF A COHERENT SHEAF, WITH THE EXPECTED DIMENSION SHIFT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The author opens the delivered proof with a sentence kept verbatim in the text (source lines 15211-15213): "The proof is exactly the same is the proof of Lemma \ref{lemma-pullback-coherent} and we suggest the reader skip it." The argument itself is then written out, namely the dimension input, the reduction of part (1) to part (2) and the full coefficient computation for part (2).

Editorial note. Canonical location of the counted proposition: Stacks tag 0FVI (sources/stacks-chow/pinned-original-source.tex line 15193).

Allowed standard prerequisites (editorial).

- The base change situation for the cycle theory and the associated dimension function delta'. Same chapter, Situation labelled situation-setup-base-change (source line 15123), invoked in the statement. Fixes the constant c, the morphism S' -> S, the base change X' -> S' and the function delta' with respect to which all dimensions are measured.
- Dimension of a base change and passage between a closed subscheme and its structure sheaf. Same chapter, Lemma labelled lemma-dimension-base-change (source line 15139) and Lemma labelled lemma-cycle-closed-coherent (source line 1776), invoked at proof lines 15216-15219. Supply the dimension bound dim_{delta'}(Z') <= k + c and the derivation of the closed subscheme statement from the coherent sheaf statement.
- Length of a pullback module along a flat local map. Algebra, Lemma labelled algebra-lemma-pullback-module (Stacks tag 02M1), invoked at proof line 15251. Supplies the equality length_B(B tensor_A M) = length_A(M) length_B(B/m_AB) whose two sides are the two coefficients computed in the counted span.
- Local Noetherianity of X and X' and coherence of the base change. Cohomology of Schemes, Lemma labelled coherent-lemma-coherent-Noetherian (Stacks tag 01XZ) and Modules, Lemma labelled modules-lemma-pullback-finite-type (Stacks tag 01B6), invoked at proof lines 15223-15228. Supply that F' is coherent so that the cycle [F']_{k+c} makes sense.

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Source reference. Original definition 0FVG: sources/stacks-chow/pinned-original-source.tex, source lines 15123–15123. Same-chapter pointer into the author environment situation at source line 15123; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02RE: sources/stacks-chow/pinned-original-source.tex, source lines 2331–2331. Same-chapter pointer into the author environment lemma at source line 2331; invoked at proof line 15212 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FVH: sources/stacks-chow/pinned-original-source.tex, source lines 15139–15139. Same-chapter pointer into the author environment lemma at source line 15139; invoked at proof line 15217 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02QY: sources/stacks-chow/pinned-original-source.tex, source lines 1776–1776. Same-chapter pointer into the author environment lemma at source line 1776; invoked at proof line 15218 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01XZ: sources/stacks-chow/tags.txt, source lines 2538–2538. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 15224 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01B6: sources/stacks-chow/tags.txt, source lines 1715–1715. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 15226 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02QW: sources/stacks-chow/pinned-original-source.tex, source lines 1702–1702. Same-chapter pointer into the author environment lemma at source line 1702; invoked at proof line 15240 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02M1: sources/stacks-chow/tags.txt, source lines 3428–3428. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 15251 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Situation 0FVG let $X \to S$ locally of finite type and let $X' \to S$ be the base change by $S' \to S$.*

1. *Let $Z \subset X$ be a closed subscheme with $\dim_\delta(Z) \leq k$ and base change $Z' \subset X'$. Then we have $\dim_{\delta'}(Z')) \leq k + c$ and $[Z']_{k+c} = g^*[Z]_k$ in $Z_{k+c}(X')$.*
2. *Let $\mathcal{F}$ be a coherent sheaf on X with $\dim_\delta(Supp(\mathcal{F})) \leq k$ and base change $\mathcal{F}'$ on X'. Then we have $\dim_\delta(Supp(\mathcal{F}')) \leq k + c$ and $g^*[\mathcal{F}]_k = [\mathcal{F}']_{k+c}$ in $Z_{k+c}(X')$.*

Proof. The proof is exactly the same is the proof of Lemma 02RE and we suggest the reader skip it.

The statements on dimensions follow from Lemma 0FVH. Part (1) follows from part (2) by Lemma 02QY and the fact that the base change of the coherent module $\mathcal{O}_Z$ is $\mathcal{O}_{Z'}$.

Proof of (2). As X, X' are locally Noetherian we may apply Cohomology of Schemes, Lemma 01XZ to see that $\mathcal{F}$ is of finite type, hence $\mathcal{F}'$ is of finite type (Modules, Lemma 01B6), hence $\mathcal{F}'$ is coherent (Cohomology of Schemes, Lemma 01XZ again). Thus the lemma makes sense. Let $W \subset X$ be an integral closed subscheme of δ-dimension k, and let $W' \subset X'$ be an integral closed subscheme of δ'-dimension $k + c$ mapping into W under $X' \to X$. We have to show that the coefficient n of $[W']$ in $g^*[\mathcal{F}]_k$ agrees with the coefficient m of $[W']$ in $[\mathcal{F}']_{k+c}$. Let $\xi \in W$ and $\xi' \in W'$ be the generic points. Let $A = \mathcal{O}_{X,\xi}$, $B = \mathcal{O}_{X',\xi'}$ and set $M = \mathcal{F}_\xi$ as an A-module. (Note that M has finite length by our dimension assumptions, but we actually do not need to verify this. See Lemma 02QW.) We have $\mathcal{F}'_{\xi'} = B \otimes_A M$. Thus we see that

$$n = \text{length}_B(B \otimes_A M) \quad \text{and} \quad m = \text{length}_A(M)\text{length}_B(B/\mathfrak{m}_A B)$$

Thus the equality follows from Algebra, Lemma 02M1. □